

The Real-World Adoption of PQC in 2026: Hype vs. Reality

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The Real-World Adoption of PQC in 2026: Hype vs. Reality

In 2026, Post-Quantum Cryptography (PQC) has moved from a theoretical discipline to a practical deployment, but its adoption is far from uniform. While major web browsers have integrated PQC, server-side support and the issuance of PQC-signed certificates are still in their early stages. This creates a significant gap between the capabilities of clients and the readiness of the broader internet infrastructure.

Hybrid Key Exchange: The Current Standard

The primary method of PQC deployment today is through **hybrid key exchange**. This approach combines a classical, well-understood cryptographic algorithm (like X25519) with a new post-quantum algorithm (like ML-KEM, specifically Kyber768). The resulting key exchange is secure as long as at least one of the algorithms remains unbroken. This dual-algorithm approach provides a safety net, ensuring security against both classical and quantum adversaries. The identifier for this hybrid mechanism in protocols like TLS 1.3 is often seen as X25519Kyber768.

Key Takeaway: Hybrid deployment is not a stopgap measure but a core security strategy. It ensures that the transition to PQC does not weaken security against today's threats while simultaneously preparing for future quantum threats.

Browser vs. Server Adoption: An Uneven Landscape

Browser support for hybrid PQC is now widespread, with all major vendors enabling it by default. However, the story is different on the server side, where adoption lags significantly.

Browser PQC Support (July 2026)

Browser	PQC Key Exchange	Default Enabled	Algorithm
Chrome 124+	✓	✓	X25519Kyber768
Firefox 128+	✓	✓	X25519Kyber768

Browser	PQC Key Exchange	Default Enabled	Algorithm
Safari 18+	✓	✓	X25519Kyber768
Edge 124+	✓	✓	X25519Kyber768

In contrast, server adoption statistics paint a different picture:

- **Top 100 websites:** ~40% support hybrid PQC.
- **Top 1 Million websites:** Only ~8.6% support hybrid PQC.
- **Internal Enterprise Services:** Estimated at less than 2%.

This disparity shows that while clients are ready, a massive effort is still required to upgrade servers, load balancers, and other network infrastructure.

The Final Frontier: PQC-Signed Certificates

The final piece of the puzzle is the Public Key Infrastructure (PKI). As of mid-2026, no publicly trusted Certificate Authority (CA) issues certificates signed with PQC algorithms like ML-DSA. The current focus is on hybrid key exchange, not yet on replacing the RSA or ECC signatures on certificates themselves.

The anticipated timeline is:

- **Late 2026:** First test/pilot PQC certificates from major CAs.
- **2027:** Initial availability of publicly trusted PQC certificates.
- **2028-2030:** Broader adoption and availability.

Daily Quiz

1. What is the primary method of PQC deployment in web protocols as of 2026?
2. Why is there a significant gap between browser PQC support and overall internet adoption?
3. What is the difference between hybrid key exchange and a PQC-signed certificate?